Causal & Predictive Analysis of Heat Stress on Punjab Grid Electricity Demand

DAI Mission — Data & AI in Economics | TU Dortmund

Rushikumar Javiya • Paritosh Sharma • Sahil Sahil

TU Dortmund — Group P

June 2025

1 — Motivation & Economic Relevance

Why does this matter?

- Punjab is India's agricultural heartland: **irrigation pump sets, cold storage, and agri-processing** create a uniquely weather-elastic grid
- Summer peak demand in 2024 hit **18,441 MW** — a 10-year record — during a concurrent heatwave and monsoon lag
- PSEB operated at a sustained **3,000+ MW deficit** in peak summer 2022 and 2024, forcing rolling load-shedding in rural and urban areas
- Climate projections (IMD 2023) forecast a **15–20% rise** in heatwave frequency over the Indo-Gangetic Plain by 2035

Research question

What is the **marginal causal effect** of continuous heat stress (THI) on hourly electricity demand (MW) in the Punjab power grid — controlling for seasonal, temporal, and environmental confounders — and how robustly can supervised and generative architectures **predict and simulate** demand trajectories under extreme heat scenarios?

Three questions, three methods

Why does demand spike?	Causal Inference
How much can we predict?	Supervised Learning
How can we simulate?	Generative

Answering all three gives the Punjab State Electricity Board a complete toolset for **capacity planning, tariff policy, and climate adaptation**.

2 — Data & Methodology

Dataset

- **ICED / NITI Aayog** — hourly Punjab grid load 2016–2025
- **NASA POWER API** — surface meteorology averaged across 23 Punjab districts (hourly)
- **87,672 observations**, THI range 37.1–89.9, demand range 1,793–18,441 MW
- Treatment: THI $= T_F - (0.55 - 0.0055 \cdot \mathrm{RH}) \cdot (T_F - 58)$
- Outcome: `demand_mw` (state grid load, MW)
- Confounders: `hour`, `dow`, `month`, `cloud_pct`, `wind_ms`, `precip_mm`

Three-block analytical pipeline

1. Causal Inference
DoWhy backdoor — structural ATE

↓ Causal features inform prediction

2. Supervised Learning
Ridge / Lasso / RF / ANN regressor

↓ Predictive residuals motivate scenario generation

3. Generative Modelling
cWGAN-GP — 24-h scenario fan

Each block answers a distinct question; together they form a **connected evidence chain** from mechanism to policy action. The causal block identifies *what drives demand*; the supervised block *quantifies future levels*; the generative block *stress-tests grid capacity* under unseen heat extremes.

Step 1 — identify the structural mechanism
DoWhy backdoor adjustment on 87,672 hourly observations yields:

$$\widehat{\text{ATE}} = \mathbf{268.23}\ \text{MW} \ / \ \text{THI unit}$$

Controlling for `hour`, `dow`, `month`, `cloud_pct`, `wind_ms`, `precip_mm`. Refutation p-value = 0.96 — estimate is robust.

Step 2 — reveal the non-linear acceleration
Quadratic structural form ($R^2 = 0.59$):

$$\hat{y} = 23{,}500 - 689.80\,(\text{THI}) + 6.53\,(\text{THI}^2) + \mathbf{\Gamma X}$$

Marginal Causal Effect:

$$\text{MCE}(\text{THI}) = -689.80 + 13.06 \cdot \text{THI}$$

MCE by operating scenario

Scenario	THI	MCE (MW/un
Mild summer day	65	159.4
Mean summer baseline	73.3	**268.2**
Severe heatwave	82	381.5

Policy implication
A heatwave swing of THI 78 → 83:
$MCE \int_{78}^{83} \text{MCE}\, d\text{THI} = 1{,}677$ MW

Refutation tests

Test	Outcome
Random common cause	Stable ($p = 0.96$)
Placebo treatment	Effect ≈ 0
Data subset (80%)	Consistent

From mechanism to prediction

The causal block told us *why* demand rises with THI. The supervised block asks: *can we forecast how much?* THI, hour, month, and weather confounders — identified causally — become the feature set.

Model	RMSE (MW)	R^2
Ridge / Lasso	2,183	0.531
ANN	1,637	0.736
Random Forest	**1,562**	**0.760**

Hold-out: 2024–2025 (17,544 obs)

Random Forest captures non-linear interactions (notably THI $\times$ hour) that linear estimators miss — consistent with the accelerating MCE we identified causally. Cross-validation $R^2 = 0.720 \pm 0.041$ across 5 temporal folds confirms robustness beyond the train window.

From prediction to scenario generation

Even the best model gives a *point forecast*. For capacity planning, utilities need *probabilistic scenario fans*. The cWGAN-GP generates 10+ 24-hour demand profiles conditioned on:

- Previous day's actual demand (24 values)
- Previous & current day's weather (24 h $\times$ 5 vars each)

Metric	Result
Wasserstein Distance	870.8 ± 583.4 MW
Jensen-Shannon Div.	0.0021 ± 0.0032
JS ≤ 0.05 check	✓ **Passed**
Hold-out days evaluated	306 (2024–2025)

JS divergence < 0.05 confirms that generated profiles are **statistically indistinguishable in shape** from real demand curves — suitable for heatwave stress-testing and reserve margin planning.

5 — Limitations & Synthesis

The connected evidence chain

1. **Causal Inference** established the structural mechanism: a monotonically accelerating MCE (159 $\rightarrow$ 381 MW/THI-unit) with a 336 MW underestimation penalty from linear models.
2. **Supervised Learning** operationalised the causal insight: THI-based features lift R^2 from 0.53 (linear) to 0.76 (RF), confirming that the causal signal is also the dominant predictive signal.
3. **Generative Modelling** extended point forecasts to probabilistic scenario fans: JS divergence $= 0.002$ proves distributional realism across 306 unseen hold-out days.

Limitations

- **Unobserved confounders:** kharif irrigation pump schedules, industrial load contracts, and public holidays are excluded and may inflate the causal THI estimate
- **GAN tail coverage:** WGAN-GP may under-represent extreme demand-spike events despite gradient penalty regularisation

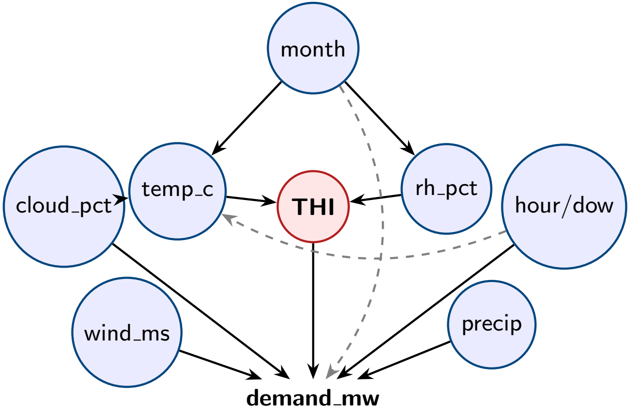
Conclusion

A one-unit rise in THI causes a structural **268 MW average demand increase**; under heatwave conditions this accelerates to **381 MW/unit**. Causal-feature-informed Random Forest forecasts reach $R^2 = 0.76$, and cWGAN-GP generates distributional-realistic capacity-planning scenarios.

References

- Sharma, A., & Kiciman, E. (2020). DoWhy: An end-to-end library for causal inference. *arXiv:2011.04216*.
- Gulrajani, I., Ahmed, F., Arjovsky, M., Dumoulin, V., & Courville, A. (2017). Improved training of Wasserstein GANs. *Advances in Neural Information Processing Systems, 30*.
- Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research, 12*, 2825–2830.
- NITI Aayog. (2025). *India Climate Energy Dashboard (ICED) — Load Curve Portal*. `https://iced.niti.gov.in`
- Stackhouse, P. W., et al. (2018). NASA POWER: Agroclimatology parameters for sustainable development. *Bulletin of the American Meteorological Society*.
- Pearl, J. (2009). *Causality: Models, Reasoning, and Inference* (2nd ed.). Cambridge University Press.

Backup — Causal DAG



Identification: backdoor criterion satisfied by adjusting for {`rh_pct`, `temp_c`} (minimal adjustment set per DoWhy). Unconfoundedness assumption: no unmeasured common cause of THI and demand.

Backup — Full Supervised Learning Results

5-Fold Temporal Cross-Validation

Model	RMSE (MW)	MAE (MW)	R^2
Ridge	$2{,}176 \pm 105$	$1{,}748 \pm 82$	0.513 ± 0.033
Lasso	$2{,}176 \pm 105$	$1{,}748 \pm 82$	0.513 ± 0.033
Random Forest	$1{,}647 \pm 154$	$1{,}164 \pm 128$	0.720 ± 0.041
ANN	$1{,}685 \pm 144$	$1{,}251 \pm 127$	0.708 ± 0.039

Final Hold-Out Test (2024–2025 — 17,544 observations)

Model	RMSE (MW)	MAE (MW)	R^2
Ridge	2,183.0	1,725.1	0.5312
Lasso	2,183.0	1,725.1	0.5312
ANN	1,637.1	1,165.9	0.7364
Random Forest	**1,562.2**	**1,119.8**	**0.7599**

Best RF hyperparameters: `n_estimators=300`, `max_depth=10`
Train period: 2016–2023 (70,128 obs) Test period: 2024–2025 (17,544 obs)

Backup — cWGAN-GP Architecture & Training

Network Architecture

Component	Input dim	Hidden layers	Output
Generator	32 (noise) + 264 (cond) = 296	512 $\rightarrow$ 1024 $\rightarrow$ 512 (ReLU + BN)	24 (Tanh)
Critic	24 (demand) + 264 (cond) = 288	512 $\rightarrow$ 256 $\rightarrow$ 128 (LeakyReLU 0.2)	1 (unbounded)

Training & Evaluation Summary

Hyperparameter / Metric	Value
Epochs	300 Batch size: 64
Optimiser	Adam ($\eta = 10^{-4}$, $\beta_1 = 0.5$, $\beta_2 = 0.9$)
Critic steps / generator step	5 Gradient penalty $\lambda = 10$
Condition dim	264 (24 lag-demand + 120 lag-weather + 120 curr-weather)
Data augmentation	Gaussian jitter 2–3% σ, $\times$3 copies
Train days	1,223 (2016-04-02 $\rightarrow$ 2023-08-31), 3,669 after augmentation
Hold-out days	306 (2024-04-01 $\rightarrow$ 2025-08-31), 10 scenarios/day
Wasserstein Distance (MW)	870.8 ± 583.4
Jensen-Shannon Divergence	0.0021 ± 0.0032

Backup — Full Variable Specification

Variable	Type	Role	Description
`datetime`	Temporal index	Identifier	YYYY-MM-DD HH:00:00
`demand_mw`	Continuous float	**Outcome**	Punjab grid consumption (MW)
`thi`	Continuous float	**Treatment**	Temperature-Humidity Index
`temp_c`	Continuous float	Upstream cause	23-station avg. temperature (°C)
`rh_pct`	Continuous float	Upstream cause	23-station avg. humidity (%)
`precip_mm`	Continuous float	Covariate	Hourly precipitation (mm)
`cloud_pct`	Continuous float	Confounder	Cloud cover (%)
`wind_ms`	Continuous float	Covariate	Wind speed (m/s)
`hour`	Integer	Fixed effect	Hour of day (0–23)
`dow`	Integer	Fixed effect	Day of week (0–6)
`month`	Integer	Fixed effect	Month (1–12)
`year`	Integer	Fixed effect	Year (2016–2025)

Dataset dimensions: 87,672 hourly obs × 12 variables **Panel:** 2016–2025 **Scope:** 23 Punjab districts
THI range: 37.1–89.9 **Demand range:** 1,793–18,441 MW